

The Open alternative for the AI world

Sergio Paniego Blanco

Machine Learning Engineer @ Hugging Face



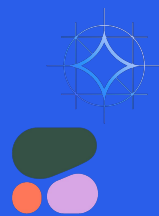
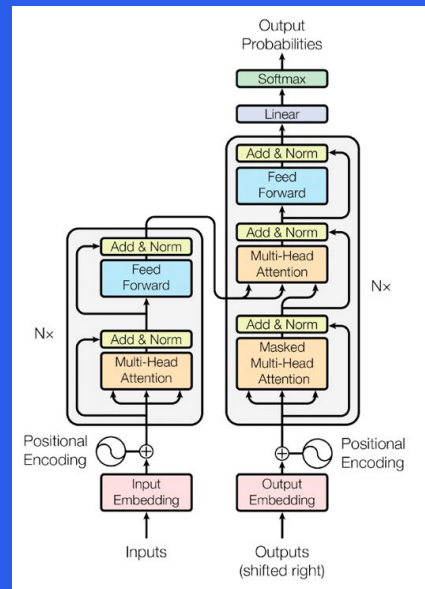
AI history 🏛️

- 1949 - Turing test
- 1956 - Dartmouth Conference: *create machines that think*
- 1958 - Perceptron
- 1986 - Backpropagation and modern neural networks
- 1997 - Deep Blue beats Kasparov
- ◈ 2006 - ImageNet, Big Data



AI boom in last decade

- 2012 - AlexNet. *Neural networks are back*
- 2017 - Transformers paper
- 2018 - BERT
- 2020 - GPT 3
- 2022 - Stable Diffusion
- 2023 - ChatGPT
- 2023-2025 - Llama, Mistral, DeepSeek, Gemma...

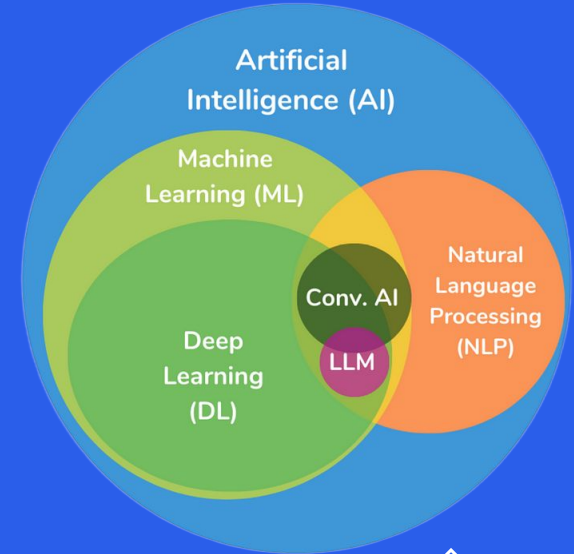


AI tasks



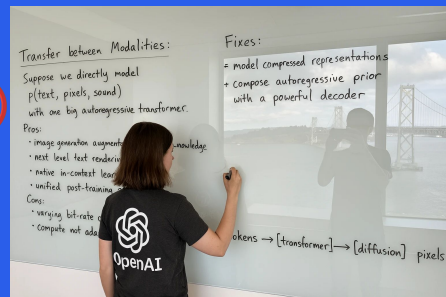
Not everything is a chatbot

- ChatGPT (text-to-text)
- Multimodality
- Agents
- Robotics
- Computer vision
- Image/video generation
- Translation
- “Simpler” tasks like classification of content (spam...)



Closed AI

- ChatGPT, Gemini, Claude, Grok...
- Veo3, Sora, Imagen...
- These models are **impressive... but closed** 
 - We can't see their code / model / prompts.
 - We don't know what data they were trained on.
 - We can't audit or replicate them.
 - We can't use them freely (licenses, limits, cost).



Why do we need open alternatives

1. Transparency & Trust



- a. Understand **how models work** and what data they were trained on.
- b. Enables **auditability** and **accountability** for ethical use.

2. Democratized Access



- a. Avoid AI being controlled by a few big players.
- b. Empower researchers, startups, and communities worldwide.

3. Reduce Dependency on Big Corporations



- a. Prevent monopolies that limit innovation and increase costs.
- b. Encourage competition and diversity in AI development.

4. Security & Ethics



- a. Open models allow **independent security reviews** to detect risks.
- b. Promote ethical AI development and responsible usage.

Open AI alternative ecosystem

- Open models
 - Llama, Gemma, Mistral, DeepSeek, Stable Diffusion...
- Open data
 - Wikipedia, Human preference, ImageNet...
- Open-source projects
 -    **Transformers** 
- Open research
 - ~~arXiv~~, peer-reviewed open papers



Gemma 3



deepseek



Meta



Who are the players?



	Mainly open	Mainly closed
Tech Giants	 Meta 	    
Other main players	     deepseek 	 

More details: [Hugging Face Heatmap](#)

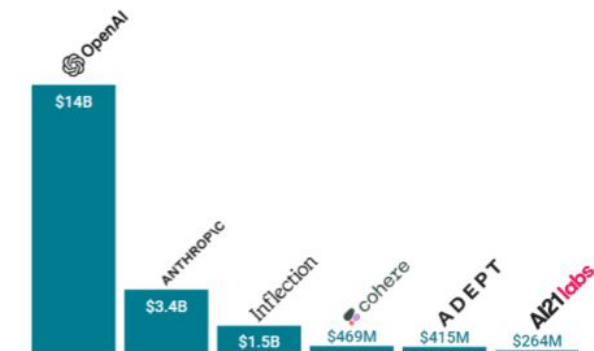
Is the market interested?



The private market is split into open vs. closed

Disclosed equity funding to LLM developers (as of 10/27/2023)

Closed-source LLMs



**Some developers may offer open-source versions of their models but keep their core models proprietary*

Open-source LLMs



**Excludes open-source developers that have not raised equity funding*

Is this actually possible?

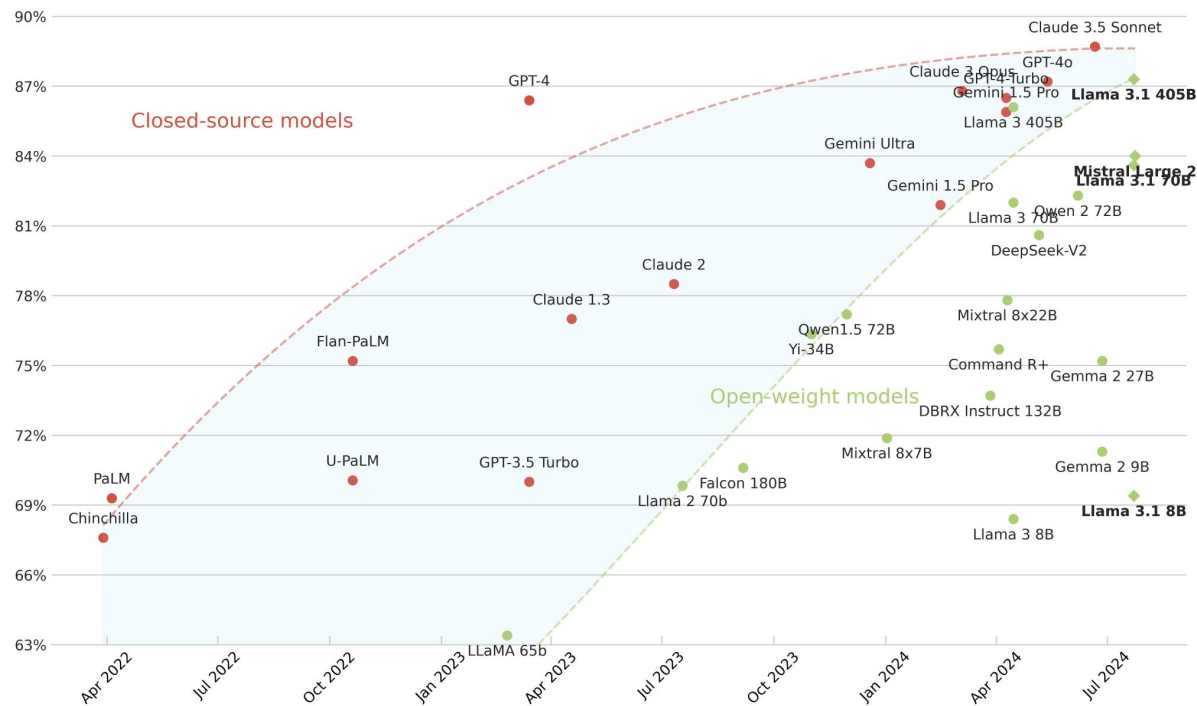


Closed-source vs. open-weight models

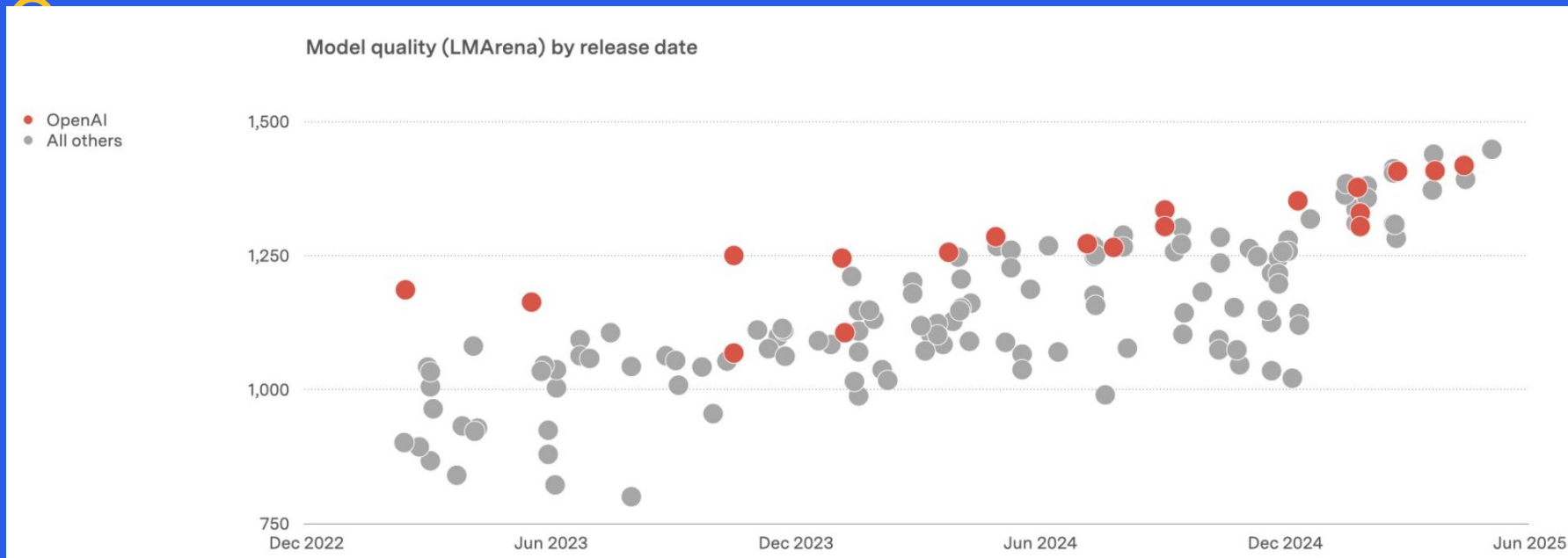
@maximelabonne

Llama 3.1 405B closes the gap with closed-source models for the first time in history.

MMLU (5-shot)

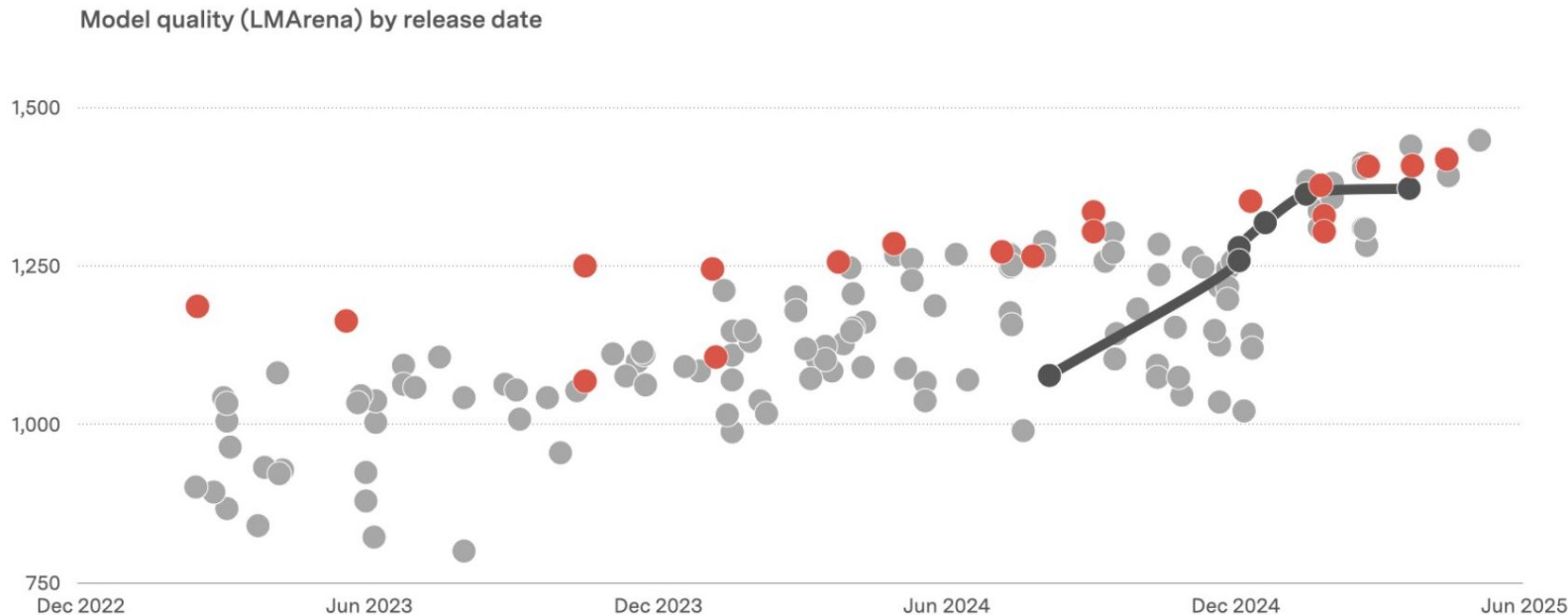


But OpenAI is just the best



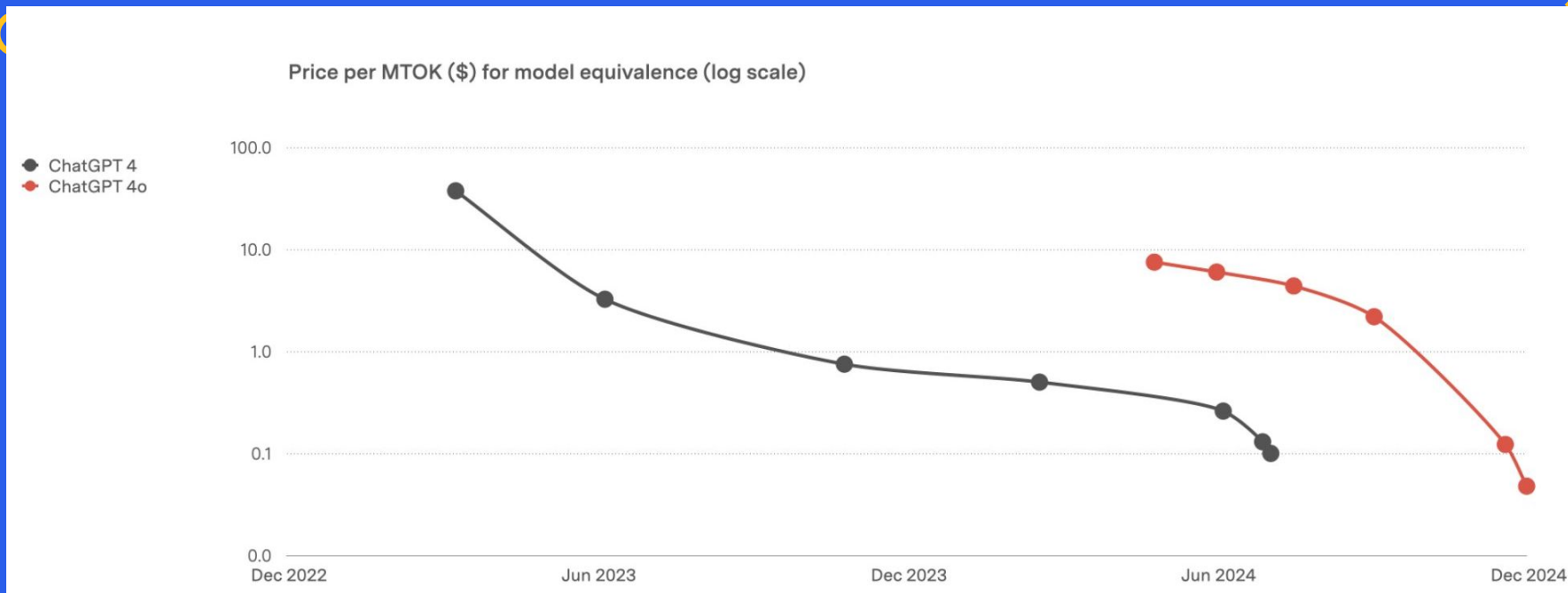
Source: [Benedict Evans](#)

But it must be luck



Source: [Benedict Evans](#)

But that must be expensive

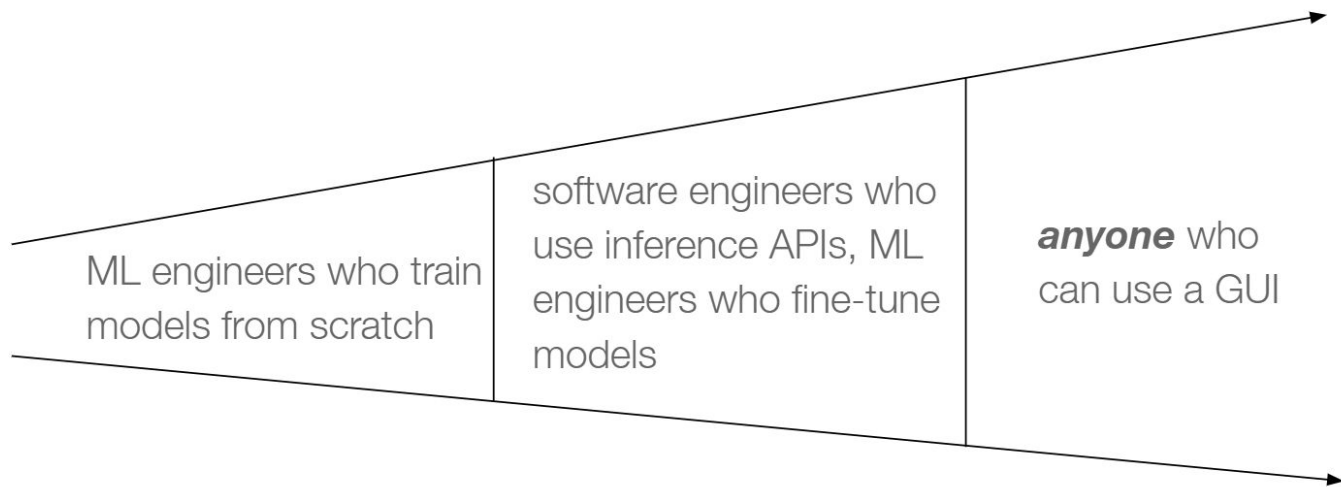


Source: [Benedict Evans](#)

HF 🤗 as motor for *Open AI*

- *Hugging Face* is born as part of this open AI alternative
- What's HF? GitHub for open AI?
 - AI open platform (+ community) where AI developers share:
 - Models (>1.5M), datasets (>300K), apps (>0.5M)...
 - Posts, blogs, courses, leaderboard, papers...
 - Set of open source projects to empower AI development and advancement:
 - 🤗 **Transformers** 🤗 **Datasets** 🤖 **smolagents**  **LeRobot** 

How HF democratizes ML/AI



DATASETS



MODELS



SPACES



Wrapping things up

- Closed AI is powerful, but limited in who it empowers. **We have to build together.**
- *Open* AI is not just an alternative, it's the foundation for **trustworthy, inclusive, and ethical innovation.**
- The open ecosystem is already catching up: **faster, cheaper, and collaborative.**
- ◆ Hugging Face, other players, and the community are making it possible.

